

Use of lower-sodium salt substitutes

WHO guideline summary



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Note

This document is a summary of a WHO guideline, the full version of which is available only in English and can be accessed at <https://www.who.int/publications/i/item/9789240105591>. This summary document does not contain all of the information found in the full guideline, but what it does contain has been extracted from the full guideline and is therefore identical to the information found in the full guideline. Anywhere in this summary document where “this guideline” or “the guideline” is used in the text, it refers to the full guideline document.

Guideline summary

Background

Globally, each year 8 million deaths are associated with poor diet. Of these, 1.9 million are attributable to high sodium intake (1). Reducing sodium intake is an effective way to reduce noncommunicable diseases (NCDs) such as cardiovascular diseases (CVDs) and chronic kidney disease by lowering blood pressure, as well as to lower risks of other conditions associated with high sodium intakes, such as gastric cancer (2).

In 2012, the World Health Organization (WHO) issued guidance on limiting sodium intake to below 2 grams per day (g/day) to reduce blood pressure and risk of CVDs (3). Member States agreed on a global target to reduce mean population sodium intake by 30% by 2025 for the prevention and control of NCDs (4) (the timeline was extended to 2030 by a World Health Assembly decision in 2019) (5). Despite efforts made by Member States, progress has been slow. The mean global sodium intake remains high – estimated to be 4.3 g/day (range 2 to 7 g/day) in 2019 (6, 7), which is more than double the WHO recommendation and demands urgent and accelerated actions.

Sodium chloride (NaCl) is the most common form of salt added to foods, both by consumers and in food manufacturing. Lower-sodium salt substitutes (LSSS) are alternatives to regular salt¹ both for discretionary use as salt added to foods by the consumer during cooking or when eating and for non-discretionary use as an ingredient present in manufactured foods and foods served at restaurants and other out-of-home settings. LSSS are also used in sodium-containing condiments, such as soy sauce and fish sauce, that are common discretionary sources of dietary sodium in some countries. These alternative salts contain less sodium than regular salt and often include potassium chloride (KCl), with or without other agents, to achieve a flavour similar to regular salt. The replacement of some of the NaCl by KCl may provide advantages, compared with regular salt, in addition to the sodium-lowering effect (9, 10), because WHO recommends increasing potassium intake from food sources to reduce blood pressure and risk of CVDs (11). That recommendation does not include obtaining potassium from supplements in tablet form or LSSS to increase potassium intake. The use of LSSS is increasingly considered by national health authorities and public health organizations as a potential sodium reduction strategy to lower blood pressure and CVD risk (12), and their use is on the rise. However, global guidance on the use of these substitutes is currently lacking. Concerns have been raised about the safety of LSSS that contain potassium, because too high a level of blood potassium (hyperkalaemia) may be harmful, especially to individuals with impaired kidney function. Therefore, it is important to systematically review existing evidence on the health effects of LSSS intake and issue WHO guidance on LSSS use through the current WHO guideline development process.

Objective

The objective of this guideline is to provide guidance on LSSS use for policy-makers, programme managers, health professionals and other stakeholders in their efforts to reduce sodium intake and reduce the risk of hypertension and related NCDs through a range of policy actions and public health interventions.

¹ “Regular salt” or “table salt” in this guideline refers to food-grade salt as defined by the *Codex standard 150-1995: Standard for food grade salt* (8). Regular table salt is regular salt that an individual adds to foods during food preparation or when eating.

Methods

This guideline was developed by the WHO Nutrition Guidance Expert Advisory Group (NUGAG) Subgroup on Diet and Health following the WHO guideline development process as outlined in the *WHO handbook for guideline development (2nd edition)* (13). This process includes a review of systematically gathered evidence by an international, multidisciplinary group of experts; assessment of the certainty of that evidence via the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework²; and consideration of additional contextual factors³ when making decisions and translating the evidence into recommendations. A group of external experts reviewed the guideline, and feedback was solicited from interested stakeholders during a public consultation.

The evidence

Evidence from the systematic review of 26 randomized controlled trials (RCTs) (published between 1986 and 2021) involving 34 961 adults and 92 children was reviewed (14). No eligible prospective cohort studies were identified. Most RCTs included some people with hypertension, and the largest RCT included only people with an elevated risk of stroke. All trials included in the systematic review specifically excluded participants in whom an increased intake of potassium could potentially cause harm – for example, people with kidney disease, impaired renal function or those using potassium-sparing medications. Some of the trials also excluded people with type 1 or 2 diabetes mellitus. LSSS interventions of any type or duration were included, provided they aimed to replace the dietary intake of any amount of sodium with another mineral or compound, although the majority of trials included in the systematic review (23 of 26 RCTs) investigated the effects of LSSS that replaced sodium with potassium. The NaCl and KCl contents of those LSSS ranged from 41 to 75% and 19 to 50%, respectively.

In adults, assignment to use LSSS compared to regular salt resulted in reductions in diastolic and systolic blood pressure during follow-up periods ranging in length from 56 days to 5 years. The mean reductions were 2.43 mmHg (95% confidence interval [CI] 3.50 lower to 1.36 lower) for diastolic blood pressure (DBP) and 4.76 mmHg (95% CI 6.01 lower to 3.50 lower) for systolic blood pressure (SBP) (*moderate* certainty evidence). The use of LSSS when compared to regular salt showed reductions in risks of:

- non-fatal stroke – risk ratio 0.90 (95% CI 0.80 to 1.01); absolute effect (AE) 20 fewer per 100 000 persons (95% CI 40 fewer to 2 more);
- non-fatal acute coronary syndrome – rate ratio 0.70 (95% CI 0.52 to 0.94); AE 150 fewer per 100 000 person-years (95% CI 250 fewer to 30 fewer); and
- cardiovascular death – rate ratio 0.77 (95% CI 0.66 to 1.00); AE 180 fewer per 100 000 person-years (95% CI 310 fewer to 0 fewer) (*moderate* certainty evidence).

Using LSSS instead of regular salt led to a mean increase of 0.12 mmol/L (95% CI 0.07 higher to 0.18 higher) in the level of potassium in the blood (*moderate* certainty evidence). LSSS compared to regular salt resulted in little to no difference in hyperkalaemia (*moderate* certainty evidence). LSSS compared to regular salt resulted in little to no difference in hypertension prevalence (*low* certainty evidence). No conclusions could be drawn about effects of LSSS on achieving blood pressure threshold or “control” as prespecified by study authors (hereafter referred to as “blood pressure control”), various other heart disease events (e.g. angina, cardiovascular symptoms), death caused by stroke, lower-than-normal blood potassium (hypokalaemia), or other adverse events (*very low* certainty evidence).

The meta-analysis of the effects of LSSS compared to regular salt or no intervention on 24 hour (h) urinary sodium excretion showed considerable heterogeneity. Despite the considerable heterogeneity, the pooled mean difference of the 11 RCTs (–19.98 mmol [–459 mg] sodium/24 h, 95% CI –35.90 to –4.06 mmol/24 h [–825

² <https://www.gradeworkinggroup.org/>

³ These include desirable and undesirable effects of the intervention, priority of the problem that the recommendation addresses, values and preferences related to the recommendation in different settings, the cost of the options available to public health officials and programme managers in different settings, feasibility and acceptability of implementing the recommendation in different settings, and the potential impact on equity and human rights.

to -93 mg/24 h], $I^2 = 91\%$, 3885 participants, 11 RCTs)⁴ was indicative of a reduction in 24 h sodium excretion on average. There was also substantial heterogeneity in the sizes of the effects of LSSS on 24 h urinary potassium excretion. The pooled mean difference of the 11 RCTs (11.44 mmol [450 mg] potassium/24 h, 95% CI 7.62 to 15.26 mmol/24 h [298 to 597 mg/24 h], $I^2 = 82\%$, 3885 participants, 11 RCTs) indicated that use of LSSS resulted in an increase in 24 h potassium excretion on average. Subgrouping by baseline 24 h urinary sodium excretion and the baseline 24 h urinary potassium excretion did not suggest important differences in the average effects of LSSS on change in DBP and change in SBP between subgroups.

There was only one study in children (3 to 17 years old) ($n = 92$) looking at the effect of LSSS instead of regular salt on DBP and SBP. No conclusions could be drawn about the effects of LSSS on DBP and SBP in children (*very low* certainty evidence). No studies included in the systematic review reported on the effect of LSSS on hypertension, blood pressure control, blood potassium, hyperkalaemia or hypokalaemia in children.

For pregnant women, no studies of the effects of LSSS on DBP and SBP, hypertension, blood pressure control, blood potassium, hyperkalaemia or hypokalaemia were found. Thus, no conclusions could be drawn about LSSS effects in pregnant women.

Most trials assessed discretionary use (i.e. salt that an individual adds to foods during food preparation or when eating) of LSSS.

In addition to the systematic review, a contextual factor narrative review was conducted (15). The review looked at additional, contextual factors related to the implementation of LSSS use. The summary of the contextual review is as follows:

- **Priority of the problem:** The global burden of disease ascribed to high blood pressure and CVDs is substantial.
- **Values and preferences:** Many studies, mainly in low- and middle-income countries, found that most people consider hypertension a serious disease with potential life-threatening consequences.
- **Resource implications:** Studies found that replacing regular salt in table salt and other foods with LSSS was cost-effective. LSSS are 1.7 times more expensive than regular salt based on the median, but overall salt is a low-cost food commodity.
- **Equity and human rights:** Lower-income or less-educated individuals were less likely to use LSSS. The higher price of LSSS was also a barrier.
- **Acceptability:** Moderate uptake of LSSS was observed for discretionary use of LSSS.
- **Feasibility:** Implementation of LSSS would be feasible overall, but several potential barriers to widespread implementation of LSSS were identified. Main barriers for consumers include: limited availability of LSSS, higher price, lack of awareness, bad taste and lack of perceived health benefit. The higher cost of LSSS and the concerns around the potentially increased risk of hyperkalaemia in those with kidney disease are potential barriers preventing governments from promoting LSSS.

Interpreting the evidence

Several observations were made in interpreting the results of the systematic review, some based directly on data from the review, and others supported by background questions and information that helped to establish the context for the recommendation (13). They are summarized below.

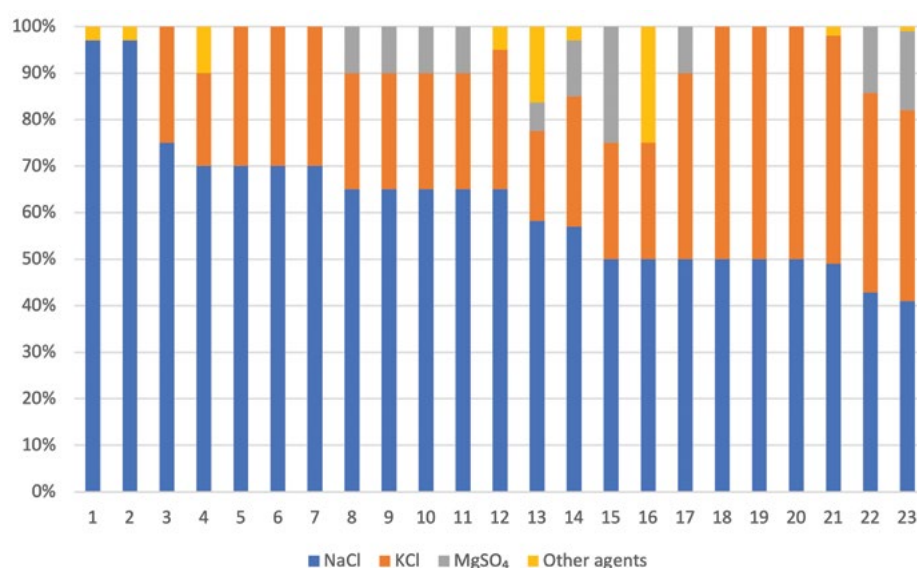
1. Varied types of LSSS were used in the RCTs

The majority of trials included in the systematic review (23 of 26) investigated the effects of LSSS that contain potassium. Since firm conclusions could not be drawn about the effects and safety of LSSS that do not replace sodium with potassium, the recommendation in this guideline applies to LSSS that contain KCl. Effectiveness and safety might be expected to vary depending on the composition of the LSSS used. NaCl

⁴ The pooled effect was not included in the published systematic review because of the I^2 threshold required by the review protocol. However, it was presented to the Nutrition Guidance Expert Advisory Group to aid assessing the pooled effect and summary statistics. GRADE was not applied to the urinary sodium and potassium outcomes; thus, the finding is not presented with certainty assessment.

and KCl contents of LSSS that contain potassium ranged from 41% to 75% and from 19% to 50%, respectively. Subgrouping by the type of LSSS was conducted for a number of outcomes based on proportion of KCl: $\geq 30\%$ KCl versus $< 30\%$ KCl versus unknown versus potassium-free LSSS. The subgroup analyses suggested no differences in average effects, although findings from the subgroup analyses in this systematic review may not all be sufficiently robust and should be interpreted with caution. In the review, subgroup analyses were often limited by very few studies or participants contributing information to certain subgroups. Some studies indicate that the blood-pressure-lowering effect of LSSS is partly due to potassium content (16). In addition to NaCl and KCl, LSSS used in the trials had various other agents, such as magnesium and calcium (see Fig. 1).

Fig. 1. Composition of LSSS used in the RCTs included in the systematic review



Note: Other agents included calcium, chitosan, folic acid, iodine, lysine and potassium citrate. One study that assessed LSSS use in bread and three studies with unknown KCl content are not included in the figure. One study appears twice in the figure above because it included two LSSS intervention arms (#18 and #22).

2. Generalizability of the evidence on safety to people at risk of high blood potassium

The systematic review found no meaningful increase in hyperkalaemia with LSSS when compared to regular salt, with little or no difference in effect for this important safety outcome at maximal follow-up of 5 years. It should be noted, however, that the evidence on hyperkalaemia presented in the review has several limitations. Very few studies reported on this important safety outcome, and studies that did report on hyperkalaemia also used variable, in some cases unclear, criteria to define the condition. Other potassium-related measures were not rigorously collected and reported. Therefore, the information on hyperkalaemia events and other potassium-related measures was unreliable. Most studies included in the review also had strict inclusion and exclusion criteria with regard to risk factors for hyperkalaemia. All included trials specifically excluded participants in whom an increased intake of potassium could cause harm – for example, people with kidney disease, impaired renal function or those using potassium-sparing medications. Only seven trials included participants judged to be at possible risk of hyperkalaemia, and four trials included participants at unclear risk of hyperkalaemia (unclear mostly because of limited reporting of the criteria assessed). Only five trials reported on this outcome, two including participants judged to be at possible risk and one including participants at unclear risk, and all information included in the meta-analysis of the systematic review came from participants judged to be at possible or unclear risk of hyperkalaemia. There was a discussion about whether the hyperkalaemia outcome should be downgraded for indirectness because the study populations have been screened and were therefore not at especially high risk of hyperkalaemia. In 75% of the trials, more than 50% of participants were hypertensive, and therefore the participants may be at moderate risk of hyperkalaemia. No downgrading for indirectness was applied for the hyperkalaemia outcome.

Finally, it should be noted that most studies that assessed safety were of limited duration and did not provide evidence of long-term safety.

Caution should be taken when applying these results directly to the general population, which is likely to include people for whom an increased intake of potassium could cause harm, and to settings where a considerable proportion of the population may have undiagnosed conditions such that increased potassium intake is potentially harmful.

3. Interpretation of the increase in blood potassium

Assignment to use LSSS instead of regular salt led to a mean increase of 0.12 mmol/L (95% CI 0.07 to 0.18) in blood potassium. To understand the clinical significance of this increase, indirect evidence exists from a systematic review and meta-analysis of RCTs that evaluated the effect on circulating potassium and renal function of increasing potassium intake with supplements (17). The reviewed trials did not include people or patients with renal impairment. The study showed that a short-term moderate increase in potassium intake using supplements (average 45 mmol or 1755 mg/day; range 22–140 mmol or 858–5460 mg/day) caused an increase in circulating potassium levels of 0.14 mmol/L (95% CI 0.09 to 0.19). This is comparable to that observed in the LSSS trials. The study also showed that potassium supplements did not seem to cause severe hyperkalaemia or deterioration in renal function in healthy people and patients whose kidney function is not impaired.

4. Lack of evidence for people without high baseline cardiovascular risk

The body of evidence involved trials that mainly restricted participants to those at high baseline cardiovascular risk, most of whom had elevated blood pressure at enrolment. Studies included only participants with hypertension (11 of 26), normal blood pressure (1 of 26), pre-hypertension (1 of 26), or included a mix of participants with and without hypertension (11 of 26). This information was unrecorded in the remaining studies (2 of 26). The largest study included only participants with an elevated risk of stroke at baseline. Due to limited data on adults without elevated blood pressure, the effect modification by hypertension status on the relationship between LSSS use and outcomes could not be reliably evaluated.

5. Generalizability of the evidence on non-discretionary use of LSSS

Most included trials investigated the implementation of LSSS as a discretionary intervention (i.e. replacement for regular table salt), which restricted the generalizability of the findings to discretionary LSSS implementation. Therefore, NUGAG was unable to draw firm conclusions about non-discretionary LSSS implementations such as their use in manufactured foods and foods sold in restaurants and other out-of-home settings. It was not possible to extrapolate the findings on discretionary LSSS use to non-discretionary use because the impacts on effectiveness and safety may vary depending on the degree and range of salt substitution as well as intakes of such foods, for which data were very limited. Moreover, considerations for the contextual factors would also differ from those associated with discretionary use of LSSS. Furthermore, the contribution of discretionary salt to total sodium intake varies considerably across countries and settings (18). Most of the included trials were conducted in China, where discretionary sources account for the majority of salt consumption. In contrast, in western countries, most salt intake comes from non-discretionary sources. This variation is an important consideration when decisions are made about the implementation of LSSS, because the absolute intakes of LSSS may vary across settings, potentially leading to variable impacts on both effectiveness and safety.

6. Children, pregnant women and household

No conclusion can be drawn about benefits and potential harms in children because there is only one trial in which children were exposed to LSSS-containing bread and outcomes were reported in children, and the results of this trial were imprecise. There is no evidence on benefits or potential harms for pregnant women. Therefore, the recommendation in this guideline does not apply to children and pregnant women.

If LSSS are used for cooking to prepare family meals, children in the home will consume LSSS. In some of the trials, the unit of allocation was the village or the household, but in these trials the exclusion criteria were applied to the whole household: if someone in the household had impaired renal function, then the

household would not participate in the trial. In trials in which households were the unit of allocation, the outcomes were only measured in adults. Because of the exclusion criteria applied to the whole household, a household where there is anyone (including children and pregnant women) with kidney disease or otherwise at high risk of hyperkalaemia should not use LSSS. Therefore, if a member in a household (including children and pregnant women) is at risk for hyperkalaemia, LSSS should not be used to prepare a family meal to be eaten by the member.

7. Additional analyses of effects of LSSS on sodium and potassium excretion or intake

Additional analyses to be included in the systematic review were requested by the NUGAG Subgroup on Diet and Health in December 2021; they were:

- ▶ addition of 24 h urinary sodium excretion as an outcome;
- ▶ addition of 24 h urinary potassium excretion as an outcome;
- ▶ subgrouping of blood pressure outcomes based on baseline 24 h urinary excretion of sodium; and
- ▶ subgrouping of blood pressure outcomes based on baseline 24 h urinary excretion of potassium.

This information was considered necessary to understand the extent to which dietary sodium intake was reduced and potassium intake was increased by using LSSS, whether the LSSS intervention was acceptable to participants, and whether and how they actually used it. Despite the heterogeneity, nine of the 11 RCTs showed a reduction in 24 h sodium excretion on average, and the pooled mean difference and its 95% CI were all indicative of a reduction. The pooled effect indicated that use of LSSS resulted in an increase in 24 h potassium excretion on average. Subgrouping by baseline 24 h urinary sodium excretion and the baseline 24 h urinary potassium excretion did not suggest differences in the average effects of LSSS on change in DBP and change in SBP between subgroups. The effects observed in the RCTs were likely due to the combination of sodium reduction and potassium increase.

8. Additional analyses of studies published after the systematic review search date

It was raised during the public consultation that a cluster-RCT (19) was published after the search date (18 August 2021) of the systematic review (14). This trial would have been eligible for inclusion in the systematic review. The crucial question was whether the new study would add data on safety in a significant manner, as the current evidence base notably lacks such data, and this lack contributes to the uncertainty about the balance between the benefits and potential harms. Additional analyses were conducted incorporating data for relevant primary outcomes reported by this study. Since there were no consequential changes to pooled effects or certainty of evidence on which the recommendation is based, the systematic review will not be formally updated for the guideline at this time.

Scope of the recommendation

Based on the review of the evidence, the scope of the recommendation was defined as follows:

The recommendation applies to discretionary use of LSSS in the form of table salt but does not apply to discretionary use of sodium-containing condiments (e.g. soy sauce, fish sauce), or non-discretionary salt already present in manufactured foods and foods served at restaurants and other out-of-home settings. The recommendation in this guideline applies to use of LSSS in which NaCl is partially replaced with KCl. The recommendation in this guideline is intended for adults in the general population and excludes individuals with kidney impairments or with other circumstances or conditions that might compromise potassium excretion (e.g. those taking potassium-sparing diuretics and potassium supplements). The recommendation does not apply to children or pregnant women. This guideline is not a clinical management guideline. Providing a recommendation on how to clinically manage and treat hypertension, kidney diseases and other conditions is beyond the scope of this guideline.

Recommendation and supporting information

Based on a review of the evidence on effects and safety, and consideration of additional contextual factors, WHO generated the following recommendation for LSSS use.

WHO recommendation

To reduce blood pressure and risk of cardiovascular diseases, WHO has recommended reducing sodium intake to less than 2 g/day (*strong* recommendation).⁵ In this context, using less regular table salt⁶ is an important part of an overall sodium reduction strategy. If choosing to use table salt, WHO suggests replacing regular table salt with lower-sodium salt substitutes that contain potassium (*conditional* recommendation).⁷ This recommendation is intended for adults (not pregnant women or children) in general populations, excluding individuals with kidney impairments or with other circumstances or conditions that might compromise potassium excretion.

This recommendation about LSSS should be aligned with the current WHO recommendations on sodium intake (3):

- ▶ WHO recommends a reduction in sodium intake to reduce blood pressure and risk of cardiovascular disease, stroke and coronary heart disease in adults (*strong* recommendation). WHO recommends a reduction to <2 g/day sodium (5 g/day salt) in adults (*strong* recommendation).
- ▶ WHO recommends a reduction in sodium intake to control⁸ blood pressure in children (*strong* recommendation). The recommended maximum level of intake of 2 g/day sodium (5 g/day salt) in adults should be adjusted downward based on the energy requirements of children relative to those of adults.

Reduction of discretionary salt intake constitutes a critical part of an overall sodium reduction strategy, especially in individuals for whom discretionary salt use is a major source of sodium intake. Importantly, the use of LSSS is only one of many means in an overall strategy to reduce sodium intake.⁹

Rationale and remarks

The following provides the reasoning behind the formulation of the recommendation (i.e. rationale) as well as remarks designed to provide context for the recommendation and facilitate its interpretation and implementation.

⁵ *Strong* recommendations are those for which the WHO guideline development group is confident that the desirable consequences of implementing the recommendation will outweigh the undesirable consequences in nearly all circumstances and can be adopted as practice or policy in most situations.

⁶ “Regular salt” or “table salt” in this guideline refers to food-grade salt as defined by the *Codex standard 150-1995: Standard for food grade salt* (8). Regular table salt is regular salt that an individual adds to foods during food preparation or when eating.

⁷ *Conditional* recommendations are those recommendations for which the WHO guideline development group is less certain that the desirable consequences of implementing the recommendation outweigh the undesirable consequences generally or in certain settings or when the anticipated net benefits are very small. Therefore, discussion may be required, including about setting-specific issues, before a *conditional* recommendation can be adopted as policy and appropriately implemented.

⁸ “Control” for this recommendation refers to the prevention of a deleterious rise in blood pressure with age.

⁹ Resolution WHA76(9) (2023) endorsed the updated menu of policy options and cost-effective interventions for reducing salt intake, such as: i) reformulation of food products to contain less salt and the setting of target levels for the amount of salt in foods and meals; ii) implementation of front-of-pack labelling and other interpretive nutrition labelling; iii) establishment of a supportive environment in public institutions such as hospitals, schools, workplaces and nursing homes, to enable lower-sodium options to be provided; iv) a behaviour change communication and mass media campaign for healthy diets; and v) implementing policies to protect children from the impact of food marketing. SHAKE Technical Package for Salt Reduction presents a suite of action menus, which is currently being updated. WHO released the second edition of the WHO global sodium benchmarks for different food categories in 2024 to assist countries set national sodium targets to reduce the sodium content of manufactured foods.

Rationale

- ▶ This recommendation is based on evidence of *moderate-to-low* certainty (an assessment of *low* certainty overall according to GRADE guidance when considering findings across all outcomes of interest) from a systematic review (14) that assessed the effects and safety of using LSSS¹⁰ compared to regular salt or no intervention. The prioritized outcomes of interest were effects on blood pressure, serum potassium (hyperkalaemia, hypokalaemia), stroke and cardiovascular events and mortality.
- ▶ The recommendation to use LSSS is based on findings from 26 RCTs in adults in which assignment to LSSS compared to regular salt resulted in reductions in DBP and SBP over 56 days to 5 years of follow-up. The mean reduction was 2.43 mmHg (95% CI 3.50 lower to 1.36 lower) for DBP and 4.76 mmHg (95% CI 6.01 lower to 3.50 lower) for SBP (*moderate* certainty evidence). The use of LSSS when compared to regular salt showed reductions in risks of non-fatal stroke (risk ratio 0.90 [95% CI 0.80 to 1.01]; AE 20 fewer per 100 000 persons [95% CI 40 fewer to 2 more]), non-fatal acute coronary syndrome (rate ratio 0.70 [95% CI 0.52 to 0.94]; AE 150 fewer per 100 000 person-years [95% CI 250 fewer to 30 fewer]) and cardiovascular death (rate ratio 0.77 [95% CI 0.66 to 1.00]; AE 180 fewer per 100 000 person-years [95% CI 310 fewer to 0 fewer]) (*moderate* certainty evidence).
- ▶ The meta-analysis of the effect of LSSS compared to regular salt or no intervention on 24 h sodium excretion showed considerable heterogeneity. Despite the considerable heterogeneity, the pooled mean difference (–19.98 mmol [–459 mg] sodium/24 h, 95% CI –35.90 to –4.06 mmol/24 h [–825 to –93 mg/24 h], $I^2 = 91\%$)¹¹ were indicative of a reduction in 24 h sodium excretion on average. There was also substantial heterogeneity in the size of the effects of LSSS on 24 h potassium excretion. The pooled mean difference (11.44 mmol [450 mg] potassium/24 h, 95% CI 7.62 to 15.26 mmol/24 h [298 to 597 mg/24 h], $I^2 = 82\%$) indicated that use of LSSS resulted in an increase in 24 h potassium excretion on average. Subgrouping by baseline 24 h urinary sodium excretion and the baseline 24 h urinary potassium excretion did not suggest important differences in the average effects of LSSS on change in DBP and change in SBP between subgroups.
- ▶ All studies in the review excluded people for whom increased potassium intake would not be advisable (e.g. those with kidney disease, those taking potassium-sparing diuretics or potassium supplements). Therefore, relevance to general populations that might include people with kidney impairments or with other circumstances or conditions that might compromise potassium excretion was uncertain. Additionally, some of the trials also excluded people with type 1 or 2 diabetes mellitus.
- ▶ In these studies, assignment to use LSSS instead of regular salt led to a mean increase of 0.12 mmol/L (95% CI 0.07 higher to 0.18 higher) in the level of potassium in the blood (*moderate* certainty evidence). LSSS compared to regular salt resulted in little to no difference in hyperkalaemia (*moderate* certainty evidence). Very few studies reported on hyperkalaemia and studies that did report on hyperkalaemia also used variable, in some cases unclear, criteria to define the condition. Other potassium-related measures presented in most of these studies were not rigorously collected and reported. Therefore, the information on hyperkalaemia events and other potassium-related measures was unreliable.

¹⁰ LSSS interventions of any type or duration were included in the systematic review, provided they aimed to replace the dietary intake of any amount of sodium with another mineral or compound, although most trials included in the systematic review (23 of 26 RCTs) investigated the effects of LSSS that replaced sodium with potassium.

¹¹ The pooled effect was not included in the published systematic review because of the I^2 threshold required by the review protocol. However, it was presented to the Nutrition Guidance Expert Advisory Group to aid assessing the pooled effect and summary statistics. The urinary sodium and potassium outcomes were not GRADED; thus, the finding is not presented with certainty assessment.

- ▶ The recommendation was assessed as *conditional* because the overall certainty of evidence was *low* according to the GRADE guidance and there was uncertainty about the balance between the benefits and potential harms, especially in settings where a considerable proportion of the population may have undiagnosed conditions for which it would not be advisable to increase potassium intakes (e.g. some low-resource settings).

Remarks

- ▶ Reducing sodium intake from both discretionary and non-discretionary use is the preferred strategy for health benefits. LSSS still contain sodium; therefore, to reduce sodium intake, the amount of sodium obtained from LSSS should be less than the amount of sodium that would have been obtained from the regular salt they replace.
- ▶ LSSS interventions of any type or duration were included in the systematic review, provided they aimed to replace the dietary intake of any amount of sodium with another mineral or compound. The recommendation statement refers to LSSS that contain potassium because most trials included in the systematic review (23 of 26 RCTs) investigated the effects of LSSS that replaced sodium with potassium, whose NaCl and KCl contents ranged from 41 to 75 % and from 19 to 50%, respectively. In addition, the blood-pressure-lowering effect of LSSS is partly due to potassium content (16); however, the percentage of KCl in the LSSS did not modify the effect in the systematic review.
- ▶ This recommendation, which applies to LSSS containing KCl, should be considered in the context of the other WHO recommendations related to potassium intake. WHO recommends consuming foods that naturally contain potassium (such as beans and peas, nuts and green vegetables) as part of a healthy diet. These foods have other nutritional benefits and should be the primary sources of dietary potassium when seeking to increase intake (11).
- ▶ This recommendation is for adults in general populations, and excludes individuals with kidney impairments or with other circumstances or conditions that might compromise potassium excretion (e.g. those taking potassium-sparing diuretics or potassium supplements). In some low-resource settings, a considerable proportion of the population may not be aware of having these conditions, and there may be individuals with undiagnosed kidney disease for whom higher potassium intakes over the long term might be of concern, and might require medical supervision and periodic assessment over time. Therefore, the use of LSSS should be implemented in settings with adequate access to health care, where conditions in which increased potassium intakes are potentially harmful (e.g. kidney disease) would not go undiagnosed for a long time. It should be noted that this guideline is not a clinical management guideline. Providing recommendations on how to clinically manage and treat hypertension, kidney diseases and other conditions is beyond the scope of this guideline.
- ▶ Discretionary salt in this recommendation is defined as salt that an individual adds to foods during cooking or when eating. It is commonly known as regular salt or table salt. This recommendation applies to discretionary use of LSSS.

- ▶ Non-discretionary salt is consumed as already present in manufactured foods and foods served at restaurants and other out-of-home settings. This recommendation does not apply to consumption of LSSS used in manufactured food products or foods sold by markets, restaurants, cafeterias and street vendors. It should also be noted that consumption of LSSS used in sodium-containing condiments, such as soy sauce and fish sauce, which are common discretionary food sources of sodium in some countries, are not included in the scope of the recommendation. There was not enough evidence on the non-discretionary consumption of LSSS (e.g. from manufactured foods) or on condiments to include them in the recommendation. However, as LSSS are increasingly used in manufactured foods and foods consumed away from home, this will alter the baseline to which discretionary LSSS are added. Such changes must be monitored at the population level to estimate total sodium and potassium intake from both discretionary and non-discretionary sources.
- ▶ There was a paucity of data for children, and no studies in children examined discretionary use of LSSS. One RCT conducted in children reported that the non-discretionary use of LSSS in bread showed little to no effect on blood pressure, but the evidence is very uncertain (*very low* certainty evidence). No studies involving pregnant women were found. Therefore, no conclusions can be drawn for children and pregnant women, and this recommendation does not apply to children and pregnant women. However, if a member in a household (including children and pregnant women) is at risk for hyperkalaemia, LSSS should not be used to prepare a family meal to be eaten by the member.
- ▶ Currently, just under half of the LSSS available globally are iodized. Action is required to ensure iodization of LSSS in order to align with national policies on salt iodization.

Translation and implementation

The recommendation in this guideline can be used by policy-makers, programme managers, health professionals and other stakeholders in their efforts to promote reduction of sodium intake and reduce the risk of hypertension and related NCDs through a range of public health policy actions and intervention programmes. Efforts should be targeted to the general population.

The recommendation in this guideline should be used in conjunction with other WHO guidance on healthy diets – in particular, guidelines relating to sodium (3) and potassium (11), as well as free sugars (20), total fat (21), saturated fatty acid and trans-fatty acids (22) and carbohydrates (23) to guide effective policy actions and intervention programmes to promote healthy diets and nutrition, and prevent diet-related NCDs.

It is important for policy-makers and programme managers to keep in mind that reducing sodium intake from both discretionary and non-discretionary use is the preferred strategy. Further efforts are needed to lower the populations' sodium intakes to below 2 g/day.

As noted in the remarks section, the use of LSSS should be implemented in settings with adequate access to health care, where kidney disease would not go undiagnosed for a long time. The recommendation can be interpreted as “Implement the recommendation to use LSSS if safety considerations can be accounted for, and monitor carefully because of potential risks of hyperkalaemia or other indicators of impaired potassium excretion that would be contraindications to LSSS use.”

It is challenging at the global level to define the settings or subpopulations where the conditions apply due to differences across countries in the way systems are set up. Conditionality gives each country the ability to assess their own situation, design and implement an adequate approach. This will require substantial discussion and involvement of various stakeholders.

Key considerations by policy-makers and programme managers when discussing the implementation of the recommendation on the LSSS use at a country level include:

- ▶ proportion of at-risk population; that is, people in the population with kidney disease or other conditions, for whom an elevated level of potassium intake should be avoided;

- ▶ demographics and trends of such at-risk populations in the country (e.g. age profile);
- ▶ access to health care (i.e. ability to detect and diagnose problems with kidney health);
- ▶ mode of use (e.g. discretionary or non-discretionary) and types of products used (e.g. table salt or manufactured foods);
- ▶ monitoring of the level of substitution of potassium in products, of sodium and potassium intakes, and of any change in the proportion of the population that is at risk;
- ▶ warning labels and health claims on products;
- ▶ consumer awareness/education on the risks and benefits of LSSS, training and meetings; and
- ▶ cost implications including higher prices of LSSS as well as costs incurred due to regulatory processes.

The recommendation applies to the discretionary use of LSSS, and therefore it is most relevant to countries where discretionary salt use is a major source of sodium intake. In populations where discretionary salt use is a small proportion of total sodium intake, measures related to discretionary LSSS will have less impact. In fact, for populations where discretionary salt use is already at very low levels, care should be taken so that consumers do not see LSSS guidance as implying that increasing their discretionary salt use with LSSS is more desirable than keeping discretionary salt use very low. Increasing discretionary salt use would increase sodium intakes. Meanwhile, some countries have already begun to implement or consider implementing the use of LSSS for non-discretionary use. This will alter the baseline to which discretionary LSSS are added. Such changes must be monitored at the population level to estimate total sodium and potassium intake from both discretionary and non-discretionary sources. For reference purposes, examples of approaches taken by some countries on the use of LSSS in manufactured foods, foods served at restaurants and other out-of-home settings and in condiments other than regular salt are provided in Annex 8 of the guideline. Although such use of LSSS is beyond the scope of this guideline, the information is provided for reference purposes.

WHO has been supporting Member States with their efforts to reduce sodium intake. In 2013, the World Health Assembly endorsed the updated menu of policy options and cost-effective interventions for reducing sodium intake, such as: i) reformulation of food products to contain less sodium and the setting of target levels for the amount of sodium in foods and meals; ii) implementation of front-of-pack labelling and other interpretive nutrition labelling; iii) establishment of a supportive environment in public institutions such as hospitals, schools, workplaces and nursing homes, to enable lower-sodium options to be provided; iv) a behaviour change communication and mass media campaign for healthy diets; and v) implementing policies to protect children from the impact of food marketing. WHO has been supporting countries to implement “Best buys and other recommended interventions for the prevention and control of noncommunicable diseases, second edition” (24). To assist countries taking policy actions, WHO released the SHAKE Technical Package for Salt Reduction in 2016 (25). The SHAKE Package is currently being updated to present a comprehensive action package for sodium reduction. Furthermore, WHO released the second edition of the WHO global sodium benchmarks for different food categories in April 2024 (26), which serves as a guide for countries in setting national sodium targets to reduce the sodium content of manufactured foods.

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